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Session : 10 MILLION SOLAR ROOFTOP SYSTEMS AND MANAGEMENT OF REVERSE POWER FLOWS ON DISTRIBUTION FEEDERS

Presented By

R.J. Vala, Chief Engineer(Tech), Gujarat Urja Vikas Nigam Ltd. (GUVNL)

Renewable Energy penetration in Gujarat

As per MNRE's Physical Progress report of state wise installed RE Capacity by 28-2-2026

Gujarat stands at

- **1st position in Renewable Installed Capacity # 44.92 GW**
- **1st position in Wind Installed Capacity # 15.20 GW**
- **2nd position in Total Solar Installed Capacity # 27.49 GW**
- **1st position in Rooftop Solar Installed Capacity # 6.67 GW**

Rooftop Solar penetration in Gujarat

- **Rooftops installed in PM SG:MBY # 5.85 Lakh/ 2.2 GW**
- **Total Rooftops installed # 12.22 Lakh/ 6.7 GW**

Residential	Industrial	Commercial	Government	Others
1172791 Nos.	17821 Nos.	16629 Nos.	9281 Nos.	6033 Nos.
4590 MW (68.5%)	1606 MW (24%)	231 MW (3.5 %)	165 MW (2.5%)	109 MW (1.5%)

Penetration of RTS in Residential sector

DISCOM	Residential Rooftops (Nos.)
DGVCL	2,15, 261
MGVCL	2,22,424
PGVCL	3,84,867
UGVCL	1,98,493
TPL-A	125755
TPL-S	25991
Total	**1172791**

District	RTS Installed	
	Nos	MW
Ahmedabad	69522	293
Surat	74940	290
Rajkot	65514	240
Vadodara	50100	200

Challenges for RE integration at local level

- Intraday variation of voltage
- Increase of Volage of LT line or DT at peak solar hours
- Operation on of a Broken conductor relay when load drops <20%
- Non-Synchronization of Inverters beyond tolerance limit

A case study of a feeder in Rajkot city

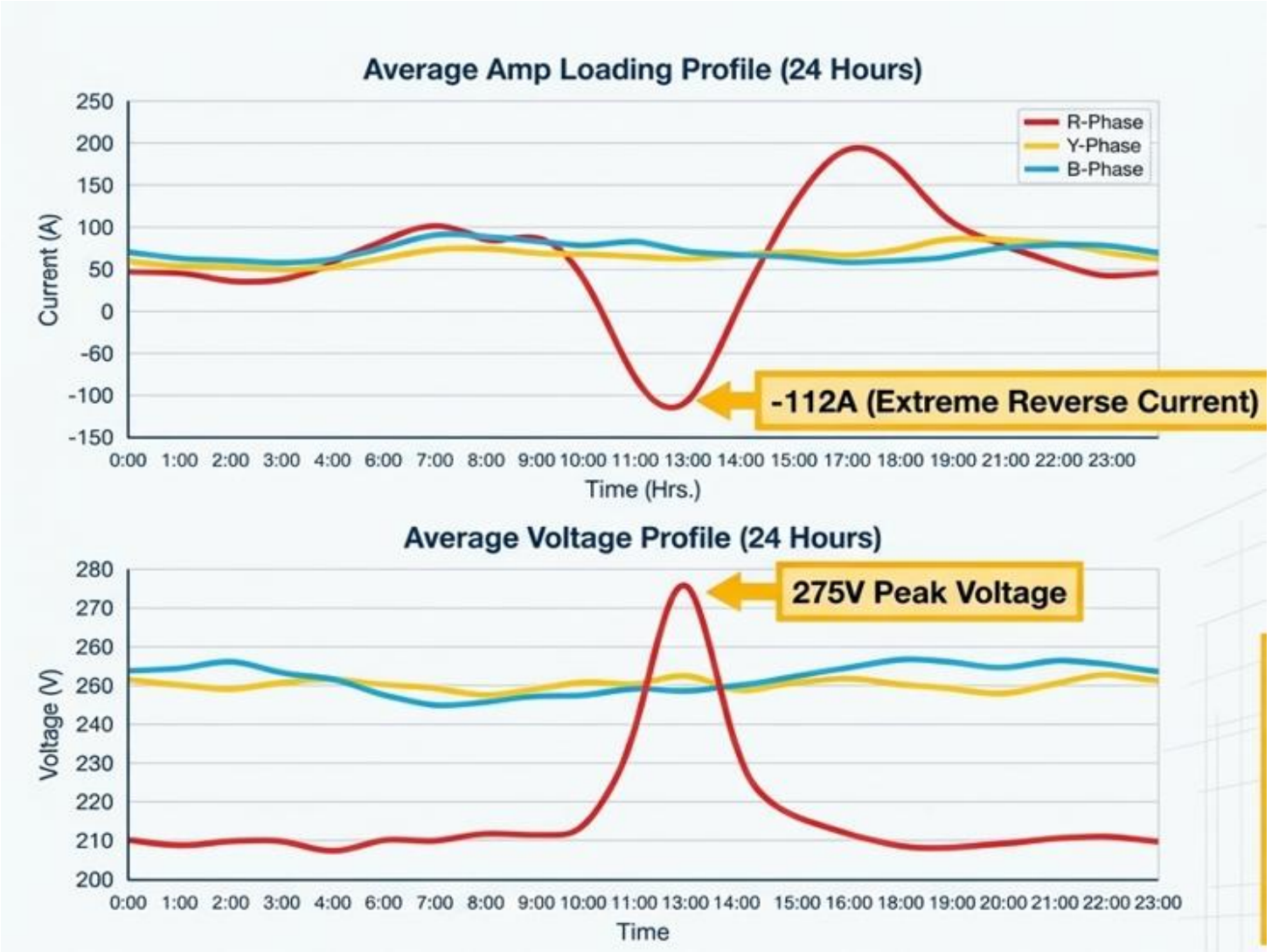
Test Sites: Laxminagar & Nana Mava feeders

Total Consumers: 2,675

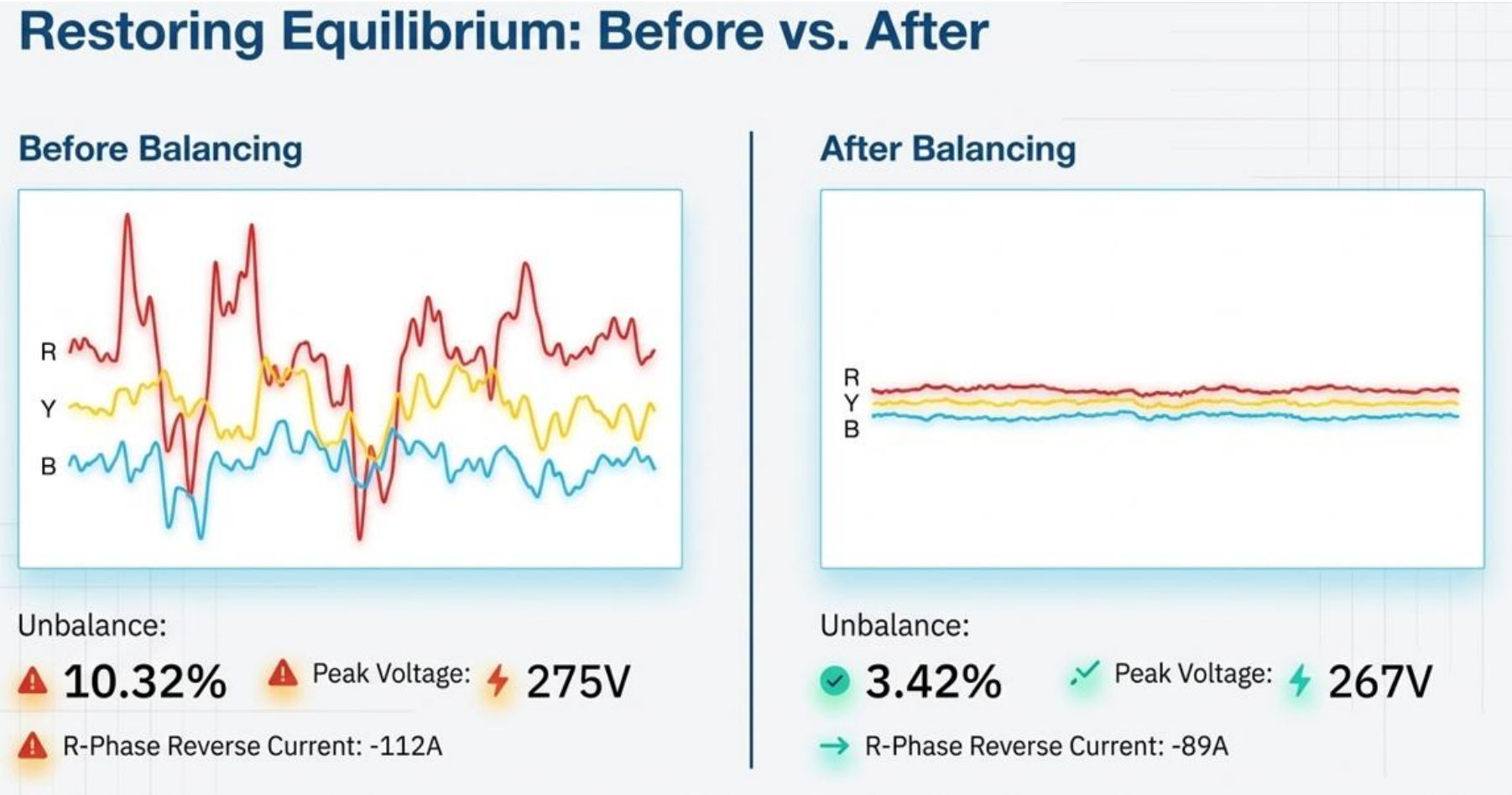
Total SRTPV Capacity: 2,906 kWp

Penetration Rate: 26.3% – 33%

Methodology: IoT modems installed at both the Distribution Transformer and consumer ends to capture 15-minute interval data.



A case study of a feeder in Rajkot city



KEY TAKEAWAYS / RECOMMENDATIONS

- Phase wise Load Balancing on LT line and DT
- TOU Discount having concession of 60 Paisa per Unit applicable for the consumption of energy during 11:00 HRS to 15:00 HRS for the consumers with smart Pre-Paid Meters
- Smart Inverters to synchronize with Grid
- 11 KV reactors to be installed on Feeders along with capacitor Bank
- Re-introduce Synchronous condenser in Grid

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)